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Effects of preconditioning with one-lung ventilation on perioperative oxygenation and oxidative stress in thoracoscopic surgery: a prospective single-center randomized controlled clinical trial

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Abstract

Background Hypoxemia is a common and clinically significant problem during one-lung ventilation (OLV). Prophylactic ventilation strategies to prevent OLV-associated hypoxemia and lung injury remain insufficiently defined.

Methods Patients scheduled for elective video-assisted thoracoscopic lung lobectomy or segmentectomy were enrolled and randomly assigned into a preconditioning group or a control group. After anesthesia induction, a visual double-lumen endotracheal tube was inserted. The preconditioning group underwent three cycles of OLV preconditioning protocol before surgery: 2 min OLV → 2 min two-lung ventilation (TLV) → 4 min OLV → 4 min TLV → 6 min OLV → 6 min TLV. The control group received conventional OLV only. Ventilation parameters: the tidal volume was set at 6 mL/kg during OLV and 8 mL/kg during TLV. Respiratory rate was adjusted to maintain an end-tidal carbon dioxide partial pressure of 35–45 mmHg. The primary outcome was the oxygenation index (arterial partial pressure of oxygen/fraction of inspired oxygen, PaO_2/FiO_2). Secondary outcomes included perioperative oxidative stress markers and the incidence of postoperative pulmonary complications (PPCs) within 7 days after surgery.

Results Seventy-four patients were included in the final analysis. The oxygenation index decreased in both groups after initiation of OLV, reaching its lowest value at 30 min. At this time point, the preconditioning group showed a significantly higher oxygenation index than the control group (209 ± 64 vs. 145 ± 43 mmHg, $P < 0.001$). No significant differences were observed between the two groups in perioperative superoxide dismutase (SOD) levels [168 ± 16 vs. 160 ± 21 U/mL, $P = 0.302$] or in the incidence of PPCs (10/38 vs. 16/36, $P = 0.105$).

Conclusions OLV preconditioning improved intraoperative oxygenation during thoracic surgery, but did not significantly affect perioperative oxidative stress or the incidence of PPCs.

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Trial registration The study was retrospectively registered in the Chinese Clinical Trial Registry on April 8, 2021, (http://www.chictr.org.cn, ChiCTR2100045203).

Keywords One-lung ventilation, Oxygenation, Oxidative stress, Postoperative pulmonary complications

Background

One-lung ventilation (OLV) is employed during thoracoscopic surgery to achieve lung isolation, thereby providing optimal surgical exposure and facilitating operative access [1]. However, OLV can lead to intrapulmonary shunting due to inadequate expansion of the ventilated lung combined with persistent perfusion of the non-ventilated lung [2]. The magnitude of this shunt is influenced by factors related to patient positioning and anesthesia management [3]. The resultant ventilation-perfusion (V/Q) mismatch compromises systemic oxygenation and end-organ metabolism [4]. Critically, the hypoxemia and potential lung injury induced by this pathophysiological process are recognized contributors to postoperative pulmonary complications (PPCs) [5].

In clinical practice, a high inhaled oxygen fraction and lung-protective ventilation (LPV) strategies are commonly used to avoid oxygen desaturation during OLV [6, 7]. Yet, the high concentration of oxygen may lead to postoperative absorptive atelectasis, potentially delaying extubation, increasing the risk of PPCs, and prolonging hospital stay [8, 9]. The LPV strategy typically involves low tidal volumes (5–6 mL/kg), positive end-expiratory pressure (PEEP) of approximately 5 cm H_2O, and periodic recruitment maneuvers [10]. The application of PEEP in lung-protective ventilation is not universally appropriate and must be individualized. In patients with bullous lung disease or chronic obstructive pulmonary disease (COPD) exhibiting significant intrinsic PEEP (auto-PEEP) and dynamic hyperinflation, the application of external PEEP must be meticulously evaluated [11, 12]. Inappropriate levels may worsen hyperinflation, elevate the risk of barotrauma, and impair venous return [10, 13–15]. Furthermore, evidence regarding the efficacy of LPV in reducing PPCs such as atelectasis and pleural effusion remains inconsistent [16, 17].

Lung injury triggered by inflammatory responses during OLV and surgery impedes postoperative recovery and is a significant concern [5, 18]. During hypoxemia, tissue levels of reactive oxygen species (ROS) increase while the activity of the antioxidant enzyme superoxide dismutase (SOD) decreases, leading to oxidative stress. This imbalance can cause lipid peroxidation, DNA damage, and protein denaturation within cellular membranes [19, 20]. Prior evidence suggests that strategies aimed at mitigating inflammatory cell infiltration, cytokine release, and oxidative stress during OLV may attenuate lung injury [18, 21].

Few studies have been published on preoperative interventions to improve oxygenation and minimize lung injury during OLV. Previous research has shown that transient, repetitive, and non-lethal hypoxia stimulation can induce cells or tissues to have a certain tolerance to hypoxia [22–25]. Therefore, we hypothesized that implementing an OLV preconditioning protocol prior to thoracic surgery would improve intraoperative oxygenation and reduce the incidence of PPCs.

Methods

Study population

This single-center, randomized controlled clinical study was approved by the Clinical Research and Laboratory Animal Ethics Committee of The First Affiliated Hospital of Sun Yat-sen University (Chairperson: Professor Chu-Rong Yan) on March 26, 2021 (Approval Number: [2021]216). It was retrospectively registered at the Chinese Clinical Trial Registry on April 8, 2021 (Registration Number: ChiCTR2100045203). This study was conducted in accordance with the ethical principles of the Declaration of Helsinki and the Consolidated Standards of Reporting Trials (CONSORT) guidelines. Patients scheduled to undergo thoracoscopic radical resection for lung cancer at our hospital from March 30, 2021, to June 30, 2022, and who had no history of psychiatric disorders, were enrolled. All participants were fully informed of the study objectives and their right to withdraw unconditionally and provided written informed consent. The original data are housed in the clinical research data repository of The First Affiliated Hospital of Sun Yat-sen University.

Inclusion criteria: age 40–70 years; American Society of Anesthesiologists (ASA) physical status I-III; preoperative cardiac function class I-II; pulmonary function forced expiratory volume in 1 s/ forced vital capacity (FEV_1/FVC) > 70%; body mass index (BMI) 17–28 kg/m^2; no history of thoracic surgery or chemoradiotherapy; no serious heart, brain, liver, or kidney diseases; no history of mental illness or long-term medication history; and the provision of signed informed consent. Exclusion criteria: A history of bronchial asthma and airway hyperresponsiveness, or preoperative anemia (hemoglobin < 9 g/dL). Drop-out criteria: Intraoperative airway pressure > 30 cm H_2O, conversion to thoracotomy, blood loss > 300 mL or requirement for blood transfusion, or OLV time < 1 h.

Group allocation

Eligible patients who provided written informed consent were randomly divided into two groups using a computer-generated random number sequence. An attending clinical anesthesiologist opened sequentially numbered, sealed, opaque envelopes containing the group assignment and implemented the corresponding ventilation strategy. The control group received direct OLV, while the study group received OLV preconditioning protocol. Patients, surgeons, nurses, and outcome assessors were blinded to the group allocation.

Study protocol for each ventilator strategy

Preconditioning Protocol: The preconditioning group underwent a specific ischemic preconditioning protocol prior to the sustained OLV for surgery. The protocol consisted of three cycles of alternating OLV and TLV as follows: 2 min of OLV → 2 min of TLV → 4 min of OLV → 4 min of TLV → 6 min of OLV → 6 min of TLV. Control Protocol: The control group received conventional OLV management without any preconditioning maneuvers. Volume-controlled ventilation was used with a fraction of inspired oxygen (FiO_2) of 0.7 and an inspiratory-to-expiratory (I: E) ratio of 1:2. During TLV, the tidal volume (Vt) was set at 8 mL/kg of predicted body weight with a respiratory rate (RR) of 12 breaths/min. During OLV, Vt was set at 6 mL/kg and RR at 14–16 breaths/min to maintain end-tidal carbon dioxide pressure ($PetCO_2$) between 35 and 45 mmHg and peak airway pressure between 15 and 30 cm H_2O. If pulse oxygen saturation (SpO_2) fell below 92%, FiO_2 was increased to 1.0 and positive end-expiratory pressure (PEEP) of 5 cm H_2O was applied. If hypoxemia persisted, bilateral lung ventilation was resumed, and the patient was excluded from the study. The study protocol and measurement time points are summarized in Fig. 1.

Anesthesia management

Total intravenous anesthesia was employed. Induction was achieved with target controlled infusion (TCI) of propofol (effect-site concentration 4.0 μg/mL), supplemented by intravenous boluses of sufentanil (0.5 μg/kg), and cisatracurium (0.2 mg/kg). After adequate anesthesia depth was confirmed, an experienced anesthesiologist placed a double-lumen endotracheal tube under video laryngoscope guidance. Correct tube position was verified and adjusted using bronchoscopic visualization. A radial artery catheter was inserted for continuous hemodynamic monitoring and arterial blood gas sampling. Anesthesia was maintained with TCI of propofol (effect-site concentration 2–3 μg/mL) and remifentanil (effect-site concentration 4–5 ng/mL). Supplemental boluses of sufentanil and cisatracurium were administered as needed. A standardized lung recruitment maneuver was performed at the conclusion of surgery. All patients received patient-controlled intravenous analgesia postoperatively. Patients were transferred to the post-anesthesia care unit (PACU) for resuscitation and were discharged from the PACU once a Steward score > 4 was achieved.

Measurements

Continuously monitored intraoperative variables included heart rate (by electrocardiography), invasive arterial blood pressure, oxygen saturation (SpO_2, by pulse oximetry), and end-tidal carbon dioxide tension ($PetCO_2$, by capnography). Patient demographics (age, sex, BMI), comorbidities (e.g., hypertension, diabetes), and lifestyle factors (smoking, alcohol use) were recorded. The primary outcome was the arterial oxygenation index (PaO_2/FiO_2 ratio) measure at five time points: before surgery (baseline, T0), 10 min after the initiation of one-lung ventilation (T1), 30 min after OLV initiation (T2), 60 min after OLV initiation (T3), and 20 min after the resumption of two-lung ventilation

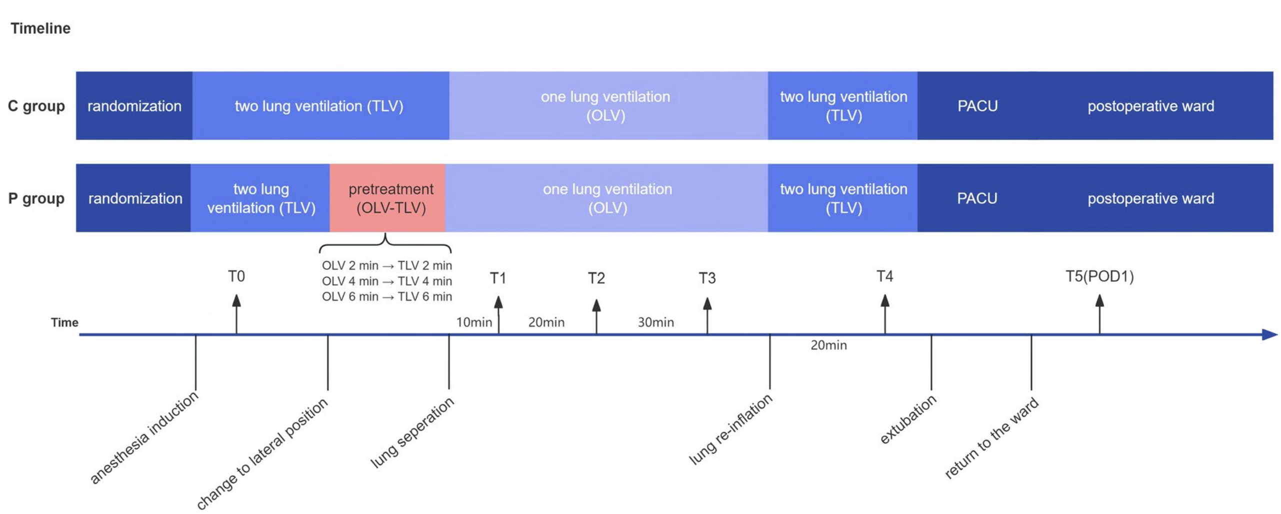

Fig. 1 Study timeline. The time points for the intervention and outcome measurements were clearly defined

(T4). Secondary outcomes included oxidative stress and inflammatory markers: superoxide dismutase (SOD), ischemia-modified albumin (IMA), and high-sensitivity C-reactive protein (Hs-CRP), measured at T0, T2, and on the first postoperative day (T5). Postoperative outcomes recorded were duration of chest tube drainage, postoperative hospital length of stay, and postoperative pulmonary complications (PPCs) within 7 days. We would explore the incidence of PPCs. PPCs were defined as the development of any of the following within the first 7 days after surgery: atelectasis, respiratory infection, pleural effusion, pneumonia, respiratory failure, or acute respiratory distress syndrome (ARDS) [26]. PPCs were assessed daily by the research team based on clinical symptoms (e.g., fever, hypoxemia, productive cough) and confirmatory diagnostic tests (e.g., chest radiographs, laboratory markers of infection). A PPC was recorded when a patient met any of the predefined diagnostic criteria.

Statistical analysis

Based on preliminary data from 10 patients demonstrating a mean PaO_2/FiO_2 ratio of 146 ± 43 mmHg during one-lung ventilation (FiO_2 = 70%) at 30 min, we performed a sample size calculation using PASS software (version 15.0). Assuming that one-lung ventilation pretreatment would improve the PaO_2 / FiO_2 ratio by 20%, with a statistical power of 80% and a significance level of 0.05, 36 patients per group were required. To accommodate potential dropouts, a total of 80 patients will be enrolled. Strategies to minimize and handle missing data include rigorous site training, automated checks, and a pre-specified analysis plan following the intention-to-treat principle with modern statistical methods and sensitivity analyses to ensure robustness. Statistical analyses were performed using SPSS 25.0 (IBM Corp., USA) for data processing and GraphPad Prism 10.0 (GraphPad Software, USA) for graphical representation. Categorical variables were compared using Chi-square or Fisher's exact tests, as appropriate, and presented as frequency (percentage). Continuous variables were assessed for normality using the Shapiro-Wilk test and homogeneity of variance using Levene's test; normally distributed variables were analyzed by independent samples t-tests (mean ± SD), while non-normally distributed variables were analyzed using the Mann-Whitney U test (median [IQR]). For longitudinal data, repeated-measures ANOVA with Holm-Bonferroni-adjusted post hoc tests

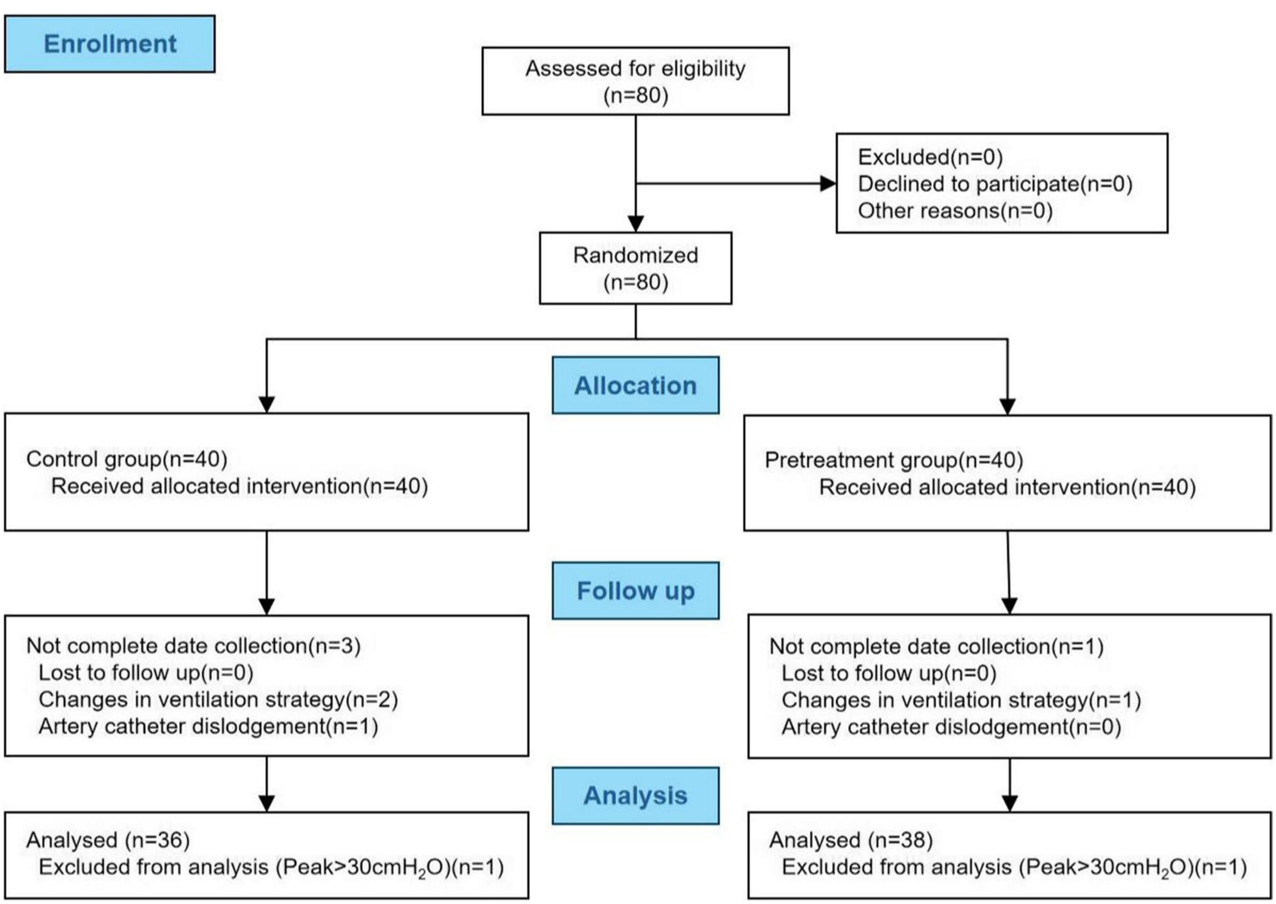

Fig. 2 Flowchart of patient enrollment and allocation according to CONSORT guidelines

was used for normally distributed data with sphericity (Mauchly's test, with Greenhouse-Geisser correction if violated), while the Friedman test with Holm-Bonferroni-adjusted pairwise comparisons was employed for non-parametric data. Statistical significance was set at $P<0.05$ (two-tailed).

Results

Patients

From March 30, 2021 to June 30, 2022, a total of 80 eligible patients were enrolled randomly allocated into two groups: a control group ($n=40$) and a study group (preconditioning group, $n=40$). Four patients in the control group were excluded: two for protocol deviation (ventilation strategy change), one for arterial catheter dislodgement, and one for intraoperative airway pressure exceeding 30 cm H_2O, as outlined in the Methods. Two patients were excluded from the preconditioning group, one each for the same two protocol deviations (ventilation strategy change and airway pressure > 30 cm H_2O). Consequently, 36 patients in the control group and 38 patients in the preconditioning group were completed the protocol and were included in the final analysis (Fig. 2). All patients underwent successful surgery and were safely transferred to the ward postoperatively.

Patient characteristics and intraoperative data

Baseline patient characteristics and surgical variables are summarized in Table 1. There were no differences in the baseline characteristics or surgical data between the two groups. As shown in Table 2, no significant

Table 1 Characteristics and pre- and post-operative data of patients

	Control group($N=36$)	Preconditioning group($N=38$)	*P*
Sex			0.337
Male	14 (38.89%)	19 (50.00%)	
Female	22 (61.11%)	19 (50.00%)	
Age (yr)	59.33 ± 7.95	56.18 ± 8.21	0.098
BMI (kg/m^2)	23.88 ± 2.48	23.11 ± 2.69	0.204
Hemoglibin (g/dL)	12.93 ± 1.68	13.04 ± 1.45	0.760
FEV_1/FVC	81.39 ± 6.46	79.97 ± 6.39	0.346
ASA			1.000
1	1 (2.78%)	1 (2.63%)	
2	33 (91.67%)	35 (92.11%)	
3	2 (5.56%)	2 (5.26%)	
Hypertension			0.737
No	25 (69.44%)	25 (65.79%)	
Yes	11 (30.56%)	13 (34.21%)	
Diabetes			0.684
No	24 (66.67%)	27 (71.05%)	
Yes	12 (33.33%)	11 (28.95%)	
Smoking			0.437
No	24 (66.67%)	22 (57.89%)	
Yes	12 (33.33%)	16 (42.11%)	
Drinking			0.670
No	33 (91.67%)	36 (94.74%)	
Yes	3 (8.33%)	2 (5.26%)	
Operation side			0.106
Left	20 (55.56%)	14 (36.84%)	
Right	16 (44.44%)	24 (63.16%)	
Operation time (min)	116.28 ± 35.02	111.58 ± 31.01	0.543
OLV time (min)	100.94 ± 27.92	99.84 ± 26.62	0.862
Duration of drain retention (day)	3.22 ± 0.76	3.45 ± 1.43	0.404
length of hospital stay (day)	5.00 ± 1.35	5.13 ± 1.61	0.706
PPCs	16 (44.4%)	10 (26.3%)	0.105
Pleural effusion	10 (27.8%)	4 (10.5%)	0.058
Atelectasis	6 (16.7%)	6 (15.8%)	0.92

Summary statistics are presented as n (%) for categorical variables and means (SD) for continuous variables

BMI body mass index, *FEV_1*/FVC Forced expiratory volume in 1 s/ Forced vital capacity, *ASA* American Society of Anesthesiologists, *OLV* one lung ventilation, *PPCs* postoperative pulmonarycomplications

Table 2 Perioperative Respiratory, Circulatory, and Arterial Blood Gas Parameters in the Pretreatment and Control Groups

		Control group ($N=36$)	Pretreatment group ($N=38$)	F (Group)/P	F (Time)/P	F (Interaction)/P
HR (bpm)	T0	74.86 ± 7.43	73.66 ± 9.83	2.121/ 0.150	79.111/ < 0.001	1.207/ 0.307
	T1	68.17 ± 6.75[a]	67.24 ± 10.89			
	T2	66.08 ± 8.55[a]	62.82 ± 10.63[a]			
	T3	63.08 ± 9.12[a]	59.76 ± 10.19[ab]			
	T4	60.94 ± 7.81[ab]	56.58 ± 10.52[ab]			
MAP (mmHg)	T0	93.97 ± 13.20	94.74 ± 13.84	4.072/ 0.047	56.061/ < 0.001	0.757/ 0.518
	T1	80.61 ± 8.71[a]	76.92 ± 10.52[a]			
	T2	73.94 ± 8.35[ab]	70.87 ± 7.48[a]			
	T3	81.67 ± 7.78[ac]	78.66 ± 8.80[ac]			
	T4	85.67 ± 9.44[ac]	91.95 ± 8.85[ac]			
$PetCO_2$ (mmHg)	T0	34.58 ± 4.26	35.84 ± 6.02	0.478/ 0.492	23.816/ < 0.001	0.919/ 0.443
	T1	37.06 ± 4.04	36.82 ± 4.42			
	T2	36.39 ± 3.71	36.00 ± 3.98			
	T3	37.14 ± 3.95[a]	37.45 ± 4.90			
	T4	31.58 ± 2.91[abcd]	32.87 ± 3.20[bcd]			
PIP (cm H_2O)	T0	15.86 ± 2.59	15.66 ± 2.18	1.522/ 0.221	73.026/ < 0.001	0.375/ 0.790
	T1	22.25 ± 4.33[a]	20.76 ± 4.60[a]			
	T2	23.14 ± 4.69[a]	22.32 ± 4.14[a]			
	T3	23.28 ± 4.12[a]	22.87 ± 4.39[a]			
	T4	16.56 ± 3.97[bcd]	16.13 ± 3.75[bcd]			
PaO2 (mmHg)	T0	329.19 ± 88.67	331.63 ± 84.37	29.734/ < 0.001	246.913/ < 0.001	3.464/ 0.024
	T1	118.75 ± 36.20[a]	171.61 ± 48.64[a*]			
	T2	98.64 ± 29.56[a]	146.55 ± 42.91[a*]			
	T3	121.33 ± 34.59[a]	181.58 ± 59.70[a*]			
	T4	272.31 ± 62.79[abcd]	311.74 ± 44.34[bcd*]			
OI(PaO_2/FiO_2) (mmHg)	T0	460.51 ± 100.84	451.25 ± 103.78	31.560/ < 0.001	264.528/ < 0.001	5.624/ 0.001
	T1	165.51 ± 52.79[a]	254.96 ± 61.92a[*]			
	T2	145.01 ± 42.61[a]	208.81 ± 63.69ab[*]			
	T3	161.53 ± 45.36[a]	233.48 ± 71.05[a*]			
	T4	394.64 ± 96.38[abcd]	429.72 ± 64.01[bcd]			
SOD (U/ml)	T0	177.06 ± 14.86	173.13 ± 13.55	1.080/ 0.302	30.869/ < 0.001	4.126/ 0.023
	T2	160.17 ± 21.39[a]	167.58 ± 16.29			
	T5	177.81 ± 19.05[b]	185.08 ± 24.30[ab]			
IMA (U/ml)	T0	81.71 ± 2.79	81.67 ± 2.98	2.659/ 0.107	28.329/ < 0.001	2.189/ 0.135
	T2	83.66 ± 3.99	82.47 ± 3.52			
	T5	86.49 ± 5.12[ab]	84.36 ± 4.32[a]			
Hs-CRP (mg/L)	T0	1.27 [0.43;2.21]	0.91 [0.48;2.14]	/	/	/
	T2	1.31 [0.51;2.14]	0.84 [0.47;2.16][*]			
	T5	139.27 [121.63;168.31] [ab]	105.45 [71.98;139.39][ab*]			

HR heart rate, *MAP* mean arterial pressure, *$PetCO_2$* end-tidal partial pressure of carbon dioxide, *PIP* peak airway pressure, *OI* oxygenation index, *PaO_2/FiO_2* Arterial partial pressure of oxygen/Fraction of inspired oxygen, *SOD* Superoxide dismutase, *IMA* Ischemia-modified albumin, *Hs-CRP* High-sensitivity C-reactive protein, *T0* before surgery, *T1* 10 min after the initiation of one-lung ventilation, *T2* 30 min after the initiation of one-lung ventilation, *T3* 60 min after the initiation of one-lung ventilation, *T4* 20 min after the resumption of two-lung ventilation. Data are presented as mean ± standard deviation

a represents $P < 0.05$ compared with T0

b represents $P < 0.05$ compared with T1

c represents $P < 0.05$ compared with T2

d represents $P < 0.05$ compared with T3

* represents $P < 0.05$ compared with the control group

differences were observed between the groups in intraoperative hemodynamic or respiratory parameters, including heart rate, mean arterial pressure, end-tidal carbon dioxide ($PetCO_2$), or peak airway pressure. No patient experienced severe bradycardia, hypotension, or subcutaneous emphysema. Consistent with prior reports, peak airway pressure increased significantly upon conversion from two-lung ventilation to one-lung ventilation (OLV).

Primary outcome

The PaO_2/FiO_2 ratio decreased significantly during OLV in both groups ($P<0.05$), with the lowest value observed at T2 (30 min after OLV). The preconditioning group maintained a significantly higher PaO_2/FiO_2 ratio at T2 compared to the control group (209 ± 64 vs. 145 ± 43 mmHg, $P<0.001$). The trend of PaO_2/FiO_2 changes across all intraoperative time points is shown in Fig. 3 and detailed in Table 2.

Changes in levels of biomarkers of oxidative stress

Perioperative changes in oxidative stress markers are shown in Fig. 4. No significant intergroup differences were found in SOD or IMA levels. Hs-CRP levels, however, increased markedly on the first postoperative day (T5), with significantly higher values in the control group compared to the preconditioning group (median [IQR]: 139 [122, 168] vs. 105 [72, 139] mg/L, $P<0.001$). Detailed statistical comparisons are provided in Table 2.

Postoperative outcomes

Postoperative outcomes are summarized in Table 1. We explored differences in the incidence of PPCs. The incidence of PPCs was numerically lower in the preconditioning group (26.3% vs. 44.4%, $P=0.105$), although this difference was not statistically significant. No significant differences were observed between the groups in chest tube duration or postoperative hospital stay.

Discussion

The incidence of hypoxemia during OLV has decreased to 5%-10% due to advances in anesthetic techniques and a better understanding of OLV physiology [4]. However, hypoxemia remains a serious concern as it increases the risk of postoperative complications such as cognitive dysfunction, atrial fibrillation, renal failure, and pulmonary hypertension, potentially leading to longer intensive care unit stays, extended hospitalization, and higher costs [27]. Therefore, preventing oxygen desaturation during thoracic surgery continues to be a priority for anesthesiologists and surgeons.

Several ventilation strategies have been developed to mitigate OLV-related lung injuries, including barotrauma, volutrauma, oxidative stress, and ischemia–reperfusion injury. Approaches such as lung-protective ventilation and continuous positive airway pressure (CPAP) to the non-ventilated lung are commonly used [6]. Animal experiments have shown that repeated systemic hypoxic preconditioning can extend survival in acute hypoxia by enhancing hypoxic tolerance [25, 28]. Therefore, we hypothesize that brief, repeated OLV preconditioning could improve oxygenation and reduce lung injury during thoracic surgery.

In this study, the PaO_2/FiO_2 ratio reached its lowest point 30 min after OLV initiation in both groups, followed by a gradual recovery. This pattern aligns with the biphasic nature of hypoxic pulmonary vasoconstriction (HPV)—an adaptive response that redistributes blood flow away from the non-ventilated lung, thereby reducing intrapulmonary shunt [29, 30]. The second phase of HPV typically peaks 30–40 min after hypoxia onset, which corresponds to the observed nadir in oxygenation [31].

Notably, the preconditioning group maintained a significantly higher PaO_2/FiO_2 ratio throughout OLV compared to the control group. The most pronounced difference was observed at T2 (208.81 ± 63.69 vs.

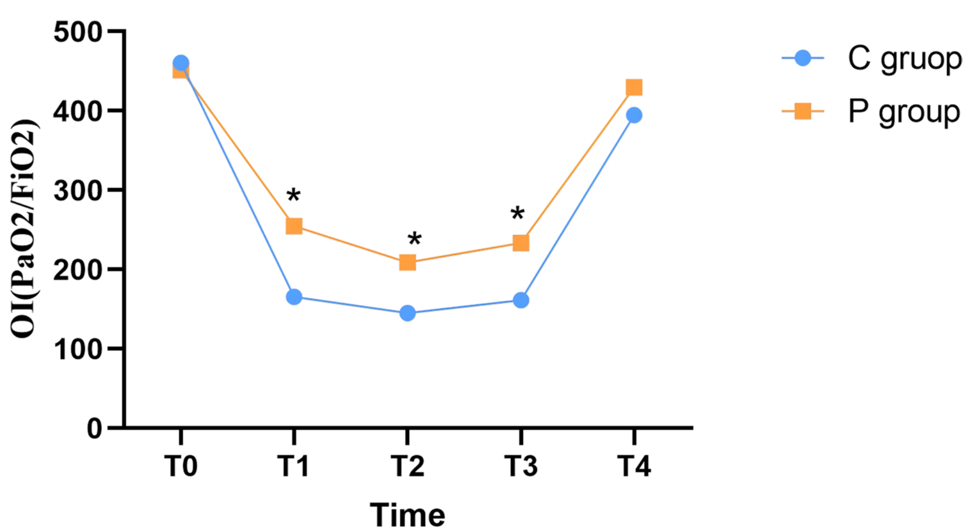

Fig. 3 Comparison of oxygenation indices between the pretreatment group and the traditional ventilation group. C group = the control group, P group = the pretreatment group, PaO2 = arterial partial pressure of oxygen, FiO2 = fraction of inspired oxygen, OI = oxygenation index, T0 = before surgery, T1 = 10 minutes after the initiation of one-lung ventilation, T2 = 30 minutes after the initiation of one-lung ventilation, T3 = 60 minutes after the initiation of one-lung ventilation, T4 = 20 minutes after the resumption of two-lung ventilation. * indicates $P<0.05$ compared with the control group

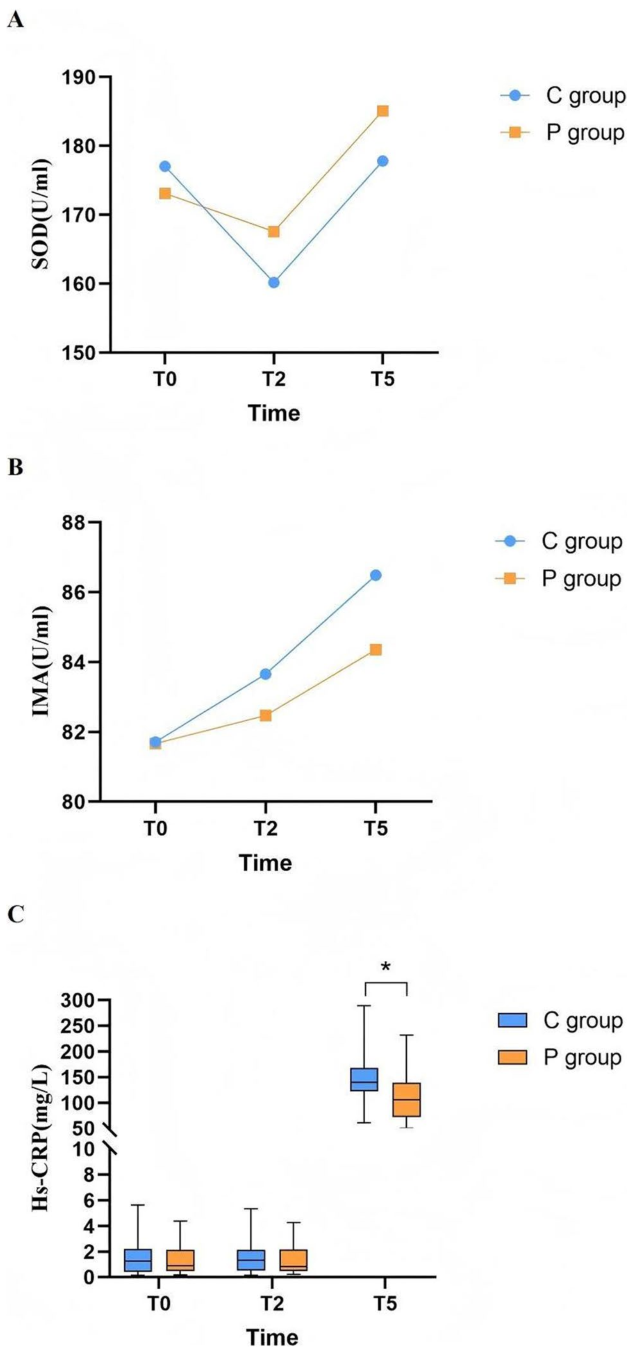

Fig. 4 Comparison of intraoperative SOD、IMA and Hs-CRP. C group = the control group, P group = the pretreatment group, SOD = superoxide dismutase, IMA = ischemia modified albumin, Hs-CRP = high-sensitivity C-reactive protein, * indicates $P < 0.05$ compared with the control group. **A**: SOD (U/ml), **B**: IMA (U/ml), **C**: Hs-CRP (mg/L)

145.01 ± 42.61 mmHg, $P < 0.001$). This improvement may be explained by enhanced HPV responsiveness through repeated, brief exposures to OLV prior to surgery, along with independent alveolar recruitment. Additionally, surgical ligation of pulmonary vessels during lobectomy may have further reduced shunt fraction by the 60-minute time point. Clinically significant hypoxemia requiring rescue measures (100% oxygen with PEEP) was uncommon, with comparable incidence between the control (2/72, 2.8%) and study (2/77, 2.6%) groups ($P = 0.94$).

Oxidative stress contributes to lung injury during OLV. SOD, a key antioxidant enzyme, decreased significantly 30 min after OLV initiation in both groups, indicating increased ROS production. Levels recovered by the first postoperative day, suggesting that OLV-induced oxidative stress is transient. IMA, a marker of ischemia and oxidative stress, did not differ between groups [32, 33]. Thus, while OLV triggers oxidative stress, our study does not provide strong evidence that preconditioning attenuates this response through reduced ROS release. In contrast, Hs-CRP a systemic inflammatory marker, was significantly lower in the preconditioning group on the first postoperative day (median: 105.45 vs. 139.27 mg/L, $P < 0.001$). This suggests that OLV preconditioning may moderate the postoperative systemic inflammatory response, potentially reducing lung injury mediated by inflammatory cytokines.

PPCs remain a major cause of morbidity after thoracic surgery [34]. It has been reported that 20% of lung resection procedures are associated with PPCs, such as acute respiratory distress syndrome (ARDS) and acute lung injury related to OLV [35]. In this study, the overall incidence of PPCs was 35%, with a numerical but non-significant reduction in the preconditioning group (26.3% vs. 44.4%, $P = 0.105$). Notably, pleural effusion was observed more frequently in the control group (10/36 vs. 4/38). The higher effusion rate in controls may reflect greater inflammatory mediator release and increased pulmonary capillary permeability during OLV. Although this difference did not reach statistical significance ($P = 0.058$), the effect size suggests a clinically relevant signal that warrants further investigation in larger trials.

This study has several limitations. First, retrospective trial registration represents a methodological deviation from prospective standards and may introduce reporting bias. Second, the single-center design limits the generalizability of our findings, and multicenter studies are needed for broader validation. Third, we did not analyze intraoperative vasopressor use, which could influence hypoxic pulmonary vasoconstriction (HPV). Fourth, the scope and timing of our oxidative stress assessment were limited, as the chosen biomarkers and sampling schedule may not have fully captured key pathways or acute dynamic changes. Fifth, although postoperative analgesia was standardized, interindividual variability in pain sensitivity and opioid requirements may have influenced inflammatory and oxidative stress markers. Finally, the study was powered only for the primary endpoint; analyses of secondary outcomes are exploratory.

In conclusion, OLV preconditioning appears to improve intraoperative oxygenation and may attenuate the systemic inflammatory response after thoracic surgery. While it did not significantly reduce the overall rate of PPCs in this cohort, it showed a promising trend toward decreasing pleural effusion. Further studies with larger sample sizes are recommended to confirm these potential benefits.

Conclusion

In summary, this study demonstrated that intermittent OLV preconditioning during thoracoscopic surgery effectively improved intraoperative oxygenation and mitigated the postoperative inflammatory response. However, this strategy did not lead to a significant reduction in either perioperative oxidative stress injury or the overall incidence of PPCs. Future research should focus on confirming these outcomes and refining the preconditioning strategy for clinical application.

Abbreviations

OLV	One-lung ventilation
TLV	Two-lung ventilation
PaO2/FiO2	Arterial partial pressure of oxygen/Fraction of inspired oxygen
PPCs	Postoperative pulmonary complications
SOD	Superoxide dismutase
LPV	Lung-protective ventilation
PEEP	Positive end-expiratory pressure
COPD	Chronic obstructive pulmonary disease
ROS	Reactive oxygen species
CONSORT	Consolidated Standards of Reporting Trials
ASA	American Society of Anesthesiologists
FEV1/FVC	Forced expiratory volume in 1 second/ Forced vital capacity
BMI	Body mass index
PetCO2	End-tidal partial pressure of carbon dioxide
TCI	Target controlled infusion
PACU	Post-anesthesia care unit
IMA	Ischemia-modified albumin
Hs-CRP	High-sensitivity C-reactive protein
CPAP	Continuous positive airway pressure
HPV	Hypoxic pulmonary vasoconstriction
ARDS	Acute respiratory distress syndrome

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Authors' contributions

J L and XH D conceived and designed the study. WZ Z and CJ W performed the experiments and collected the data. J L and DZ Z conducted the data analysis. J L and XH D wrote the manuscript. JY F revised the manuscript. All authors reviewed and approved the final manuscript.

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Data availability

The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Clinical Research and Laboratory Animal Ethics Committee of the First Affiliated Hospital of Sun Yat-sen University (Chairperson: Professor Chu-Rong Yan) on March 26, 2021 (Approval Number: [2021]216). The study has been registered at the Chinese Clinical Trial Registry on April 8, 2021 (Registration Number: ChiCTR2100045203). This study adhered to the principles of the Declaration of Helsinki. Informed consent was obtained in writing from all individuals involved.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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